

CERTIFIED ETHICAL HACKER (CEH)

Final Course Project

Project 3

Multi-Router OSPF Network

Design, Configuration, Verification & Hardening

Platform: Cisco Packet Tracer · Cisco 2811 Routers, 2960 Switches

Routing Protocol: OSPFv2 (Single Area 0)

Prepared by: Anwar

Table of Contents

Table of Contents	2
1. Executive Summary.....	3
2. Objectives	3
3. Network Topology.....	3
4. IP Addressing Scheme	4
4.1 Serial (WAN) Links — /30 Point-to-Point	4
4.2 LAN Segments and Loopbacks	4
5. Configuration Methodology	5
6. Verification and Results	5
6.1 OSPF Neighbor Adjacencies	5
6.2 OSPF Routing Table	5
6.3 End-to-End Connectivity Test	5
6.4 Path Trace (Optional)	6
7. Security Hardening — OSPF MD5 Authentication	6
8. Challenges and Resolutions	6
9. Conclusion	7
Appendix A — Full Router Configurations	8
A.1 Router R1	8
A.2 Router R2	8
A.3 Router R3	9
A.4 Router R4	10
Appendix B — Professor Defense Q&A.....	Error! Bookmark not defined.

1. Executive Summary

This report documents the design, configuration, verification, and hardening of a four-router Open Shortest Path First (OSPF) network built in Cisco Packet Tracer. Four Cisco 2811 routers are interconnected in a redundant ring using point-to-point serial links, with each router serving a dedicated LAN of two end hosts through a Cisco 2960 switch. A single OSPF backbone area (Area 0) distributes routing information across the entire topology.

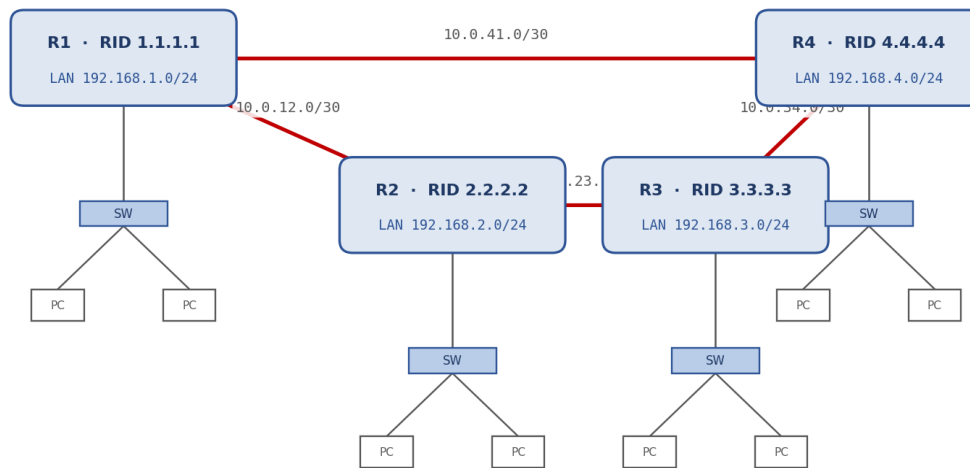
The project demonstrates full OSPF adjacency formation, correct route propagation, end-to-end inter-LAN connectivity, and the ring's inherent path redundancy. As a Certified Ethical Hacker exercise, the deployment is further hardened with MD5 neighbor authentication to defend the routing control plane against spoofed-update (LSA injection) attacks.

2. Objectives

- Design a four-router ring topology using Cisco 2811 routers and WIC-2T serial modules.
- Assign a structured IP addressing scheme using /30 serial links and /24 LANs.
- Configure OSPFv2 in a single backbone area (Area 0) on all four routers.
- Verify full adjacency, complete route distribution, and end-to-end reachability.
- Demonstrate the redundancy provided by the ring topology.
- Harden the OSPF control plane using MD5 authentication (security objective).

3. Network Topology

The four routers form a closed ring: R1–R2–R3–R4–R1. Each router connects to its two neighbors through serial interfaces and to its local LAN through FastEthernet0/0 and a 2960 switch. Figure 1 shows the logical topology and addressing.

Figure 1. Four-Router OSPF Ring Topology (Area 0)

All serial links are /30 point-to-point in OSPF area 0. Each router hosts a LAN via a 2960 switch.

Figure 1. Multi-router OSPF ring topology with per-link /30 serial subnets and per-site /24 LANs.

Device inventory: 4 × Cisco 2811 routers (each with a WIC-2T serial module), 4 × Cisco 2960 switches, and 8 × PC end hosts (two per LAN).

4. IP Addressing Scheme

4.1 Serial (WAN) Links — /30 Point-to-Point

Each serial link is a dedicated /30 subnet. The DCE end of every link (shown first) provides the clock signal.

Subnet	Local End	IP	Remote End	IP
10.0.12.0/30	R1 S0/0/0	10.0.12.1	R2 S0/0/1	10.0.12.2
10.0.23.0/30	R2 S0/0/0	10.0.23.1	R3 S0/1/0	10.0.23.2
10.0.34.0/30	R3 S0/1/1	10.0.34.1	R4 S0/0/1	10.0.34.2
10.0.41.0/30	R4 S0/0/0	10.0.41.1	R1 S0/0/1	10.0.41.2

Table 1. Serial link addressing (10.0.x.0/30 ranges).

4.2 LAN Segments and Loopbacks

Router	Gateway (Fa0/0)	Switch	Host Addresses	Router-ID (Lo0)
R1	192.168.1.1/24	Switch-1	PC1 .10 / PC2 .11	1.1.1.1
R2	192.168.2.1/24	Switch-2	PC3 .10 / PC4 .11	2.2.2.2
R3	192.168.3.1/24	Switch-3	PC5 .10 / PC6 .11	3.3.3.3
R4	192.168.4.1/24	Switch-4	PC7 .10 / PC8 .11	4.4.4.4

Table 2. LAN gateways, host addressing, and loopback router-IDs.

5. Configuration Methodology

Configuration proceeded in three phases: (1) physical build — placing devices, installing WIC-2T serial modules, and cabling the ring with serial DCE cables; (2) interface and OSPF configuration on each router; and (3) verification. A key practical finding during the build was that the WIC-2T modules were installed in different slots across routers, producing different serial interface names (Serial0/0/x on R1, R2, R4 versus Serial0/1/x on R3). Interface names were confirmed per device with `show ip interface brief` before addressing, and physical wiring was validated with `show cdp neighbors` to ensure both ends of every link shared the correct /30 subnet.

OSPF design choices: all four routers run a single process (router ospf 1) in Area 0; router-IDs are pinned to loopback addresses for stability; LAN interfaces are declared passive to suppress unnecessary hello traffic; and network statements use wildcard masks to enable OSPF precisely on the intended interfaces.

6. Verification and Results

The following evidence confirms the network is fully operational. Each figure corresponds to a command executed on the live topology.

6.1 OSPF Neighbor Adjacencies

Command: `show ip ospf neighbor` (executed on R1)

[INSERT SCREENSHOT]

Figure 2. R1 shows two neighbors — 2.2.2.2 (R2) and 4.4.4.4 (R4) — both in the FULL state, confirming adjacency around the ring.

Each router forms a FULL adjacency with both of its ring neighbors, confirming that hello parameters, subnet masks, and (after hardening) authentication keys match on every link.

6.2 OSPF Routing Table

Command: `show ip route ospf` (executed on R1)

[INSERT SCREENSHOT]

Figure 3. R1's OSPF routing table showing 'O' routes to all remote LANs (192.168.2.0, 192.168.3.0, 192.168.4.0) and loopbacks, with equal-cost dual paths to destinations reachable both ways around the ring.

The presence of two equal-cost entries for destinations such as 192.168.3.0 and 3.3.3.3 is direct evidence of the ring's redundancy: OSPF has computed a path in each direction.

6.3 End-to-End Connectivity Test

Command: `ping 192.168.3.10` (from PC1 on R1's LAN to PC5 on R3's LAN)

[INSERT SCREENSHOT]

Figure 4. Successful ping from PC1 (192.168.1.10) to PC5 (192.168.3.10) across the ring — Sent = 4, Received = 4, 0% loss. The first packet timed out only while ARP resolved.

A successful ping between hosts on non-adjacent LANs proves that OSPF has distributed reachability information correctly and that traffic is routed end-to-end across multiple routers.

6.4 Path Trace (Optional)

Command: `tracert 192.168.3.10` (from PC1)

[INSERT SCREENSHOT]

Figure 5. Traceroute from PC1 to PC5 listing each router hop taken across the OSPF ring.

7. Security Hardening — OSPF MD5 Authentication

Threat. By default, OSPF trusts any device on a connected link that can produce syntactically valid hello and Link-State Advertisement (LSA) packets. An attacker with access to a serial link (or an emulated router) could inject forged LSAs to advertise false routes, redirecting traffic through an attacker-controlled path (a man-in-the-middle) or blackholing it entirely. This is a control-plane attack that undermines the integrity of the entire routing domain.

Mitigation. Cryptographic neighbor authentication was enabled on every serial interface using MD5 message digests. Each router computes a keyed hash over its OSPF packets; a neighbor accepts an update only if it can reproduce the same digest using the shared secret key. Packets that fail the check — including any injected by an attacker without the key — are silently discarded.

Configuration applied to each serial interface:

```
interface Serial0/0/0
 ip ospf authentication message-digest
 ip ospf message-digest-key 1 md5 Cisco123
!
interface Serial0/0/1
 ip ospf authentication message-digest
 ip ospf message-digest-key 1 md5 Cisco123
```

Verification. After applying matching keys on both ends of every link, adjacencies remain FULL, confirming legitimate neighbors still communicate. A mismatched or absent key prevents adjacency — demonstrating that unauthenticated (attacker) routers are rejected.

[INSERT SCREENSHOT]

Figure 6. 'show ip ospf interface Serial0/0/0' reporting 'Message digest authentication enabled', alongside 'show ip ospf neighbor' still showing FULL adjacencies after hardening.

8. Challenges and Resolutions

Several realistic issues arose during the build and were resolved systematically:

- **Inconsistent serial interface names.** WIC-2T modules landed in different slots, so serial ports were Serial0/0/x on some routers and Serial0/1/x on others. Resolved by confirming names per device with `show ip interface brief` before configuring.
- **Subnet mismatches on links.** Because cables were connected across several attempts, some serial IPs did not match the physically wired port. Resolved by mapping the real wiring with `show cdp neighbors` and re-aligning each link's /30 addresses.
- **Overlapping-address errors during correction.** Swapping two interface IPs directly triggered '% overlaps' errors; resolved by clearing one interface (no ip address) before reassigning both.
- **Missing ring-closing link.** One serial cable had not connected to the intended ports, splitting the topology into two islands. Identified via CDP (a neighbor was absent) and fixed by re-drawing the serial DCE cable between the correct ports.

9. Conclusion

The four-router OSPF ring was successfully designed, built, and verified. All adjacencies reached the FULL state, every router learned the complete network via OSPF, and end-to-end connectivity across the ring was confirmed with a zero-loss ping. The topology's redundancy was evidenced by equal-cost dual paths in the routing table. Finally, the routing control plane was hardened with MD5 authentication, addressing a well-known OSPF security weakness and aligning the exercise with Certified Ethical Hacker defensive objectives.

Appendix A — Full Router Configurations

A.1 Router R1

```
hostname R1
no ip domain-lookup
!
interface Loopback0
 ip address 1.1.1.1 255.255.255.255
!
interface FastEthernet0/0
 ip address 192.168.1.1 255.255.255.0
 no shutdown
!
interface Serial0/0/0
 description Link to R2 (DCE)
 ip address 10.0.12.1 255.255.255.252
 ip ospf authentication message-digest
 ip ospf message-digest-key 1 md5 Cisco123
 clock rate 64000
 no shutdown
!
interface Serial0/0/1
 description Link to R4 (DTE)
 ip address 10.0.41.2 255.255.255.252
 ip ospf authentication message-digest
 ip ospf message-digest-key 1 md5 Cisco123
 no shutdown
!
router ospf 1
 router-id 1.1.1.1
 passive-interface FastEthernet0/0
 network 192.168.1.0 0.0.0.255 area 0
 network 10.0.12.0 0.0.0.3 area 0
 network 10.0.41.0 0.0.0.3 area 0
 network 1.1.1.1 0.0.0.0 area 0
!
end
```

A.2 Router R2

```
hostname R2
no ip domain-lookup
!
interface Loopback0
 ip address 2.2.2.2 255.255.255.255
!
interface FastEthernet0/0
 ip address 192.168.2.1 255.255.255.0
 no shutdown
!
interface Serial0/0/0
```



```
description Link to R3 (DCE)
ip address 10.0.23.1 255.255.255.252
ip ospf authentication message-digest
ip ospf message-digest-key 1 md5 Cisco123
clock rate 64000
no shutdown
!
interface Serial0/0/1
description Link to R1 (DTE)
ip address 10.0.12.2 255.255.255.252
ip ospf authentication message-digest
ip ospf message-digest-key 1 md5 Cisco123
no shutdown
!
router ospf 1
router-id 2.2.2.2
passive-interface FastEthernet0/0
network 192.168.2.0 0.0.0.255 area 0
network 10.0.23.0 0.0.0.3 area 0
network 10.0.12.0 0.0.0.3 area 0
network 2.2.2.2 0.0.0.0 area 0
!
end
```

A.3 Router R3

Note: R3's serial ports are Serial0/1/0 and Serial0/1/1 (WIC-2T in slot 1).

```
hostname R3
no ip domain-lookup
!
interface Loopback0
ip address 3.3.3.3 255.255.255.255
!
interface FastEthernet0/0
ip address 192.168.3.1 255.255.255.0
no shutdown
!
interface Serial0/1/1
description Link to R4 (DCE)
ip address 10.0.34.1 255.255.255.252
ip ospf authentication message-digest
ip ospf message-digest-key 1 md5 Cisco123
clock rate 64000
no shutdown
!
interface Serial0/1/0
description Link to R2 (DTE)
ip address 10.0.23.2 255.255.255.252
ip ospf authentication message-digest
ip ospf message-digest-key 1 md5 Cisco123
no shutdown
```

```
!  
router ospf 1  
  router-id 3.3.3.3  
  passive-interface FastEthernet0/0  
  network 192.168.3.0 0.0.0.255 area 0  
  network 10.0.34.0 0.0.0.3 area 0  
  network 10.0.23.0 0.0.0.3 area 0  
  network 3.3.3.3 0.0.0.0 area 0  
!  
end
```

A.4 Router R4

```
hostname R4  
no ip domain-lookup  
!  
interface Loopback0  
  ip address 4.4.4.4 255.255.255.255  
!  
interface FastEthernet0/0  
  ip address 192.168.4.1 255.255.255.0  
  no shutdown  
!  
interface Serial0/0/0  
  description Link to R1 (DCE)  
  ip address 10.0.41.1 255.255.255.252  
  ip ospf authentication message-digest  
  ip ospf message-digest-key 1 md5 Cisco123  
  clock rate 64000  
  no shutdown  
!  
interface Serial0/0/1  
  description Link to R3 (DTE)  
  ip address 10.0.34.2 255.255.255.252  
  ip ospf authentication message-digest  
  ip ospf message-digest-key 1 md5 Cisco123  
  no shutdown  
!  
router ospf 1  
  router-id 4.4.4.4  
  passive-interface FastEthernet0/0  
  network 192.168.4.0 0.0.0.255 area 0  
  network 10.0.41.0 0.0.0.3 area 0  
  network 10.0.34.0 0.0.0.3 area 0  
  network 4.4.4.4 0.0.0.0 area 0  
!  
end
```